

3DM Homework Hints and Solutions

Graph Theory Chapter 1

1.2: One way to find the size of a graph is to use Theorem 1.

1.4: Hint: look at Corollary 2.

1.5: Use proof by contradiction: Suppose, to the contrary, that there exists a nontrivial irregular graph, G . Denote the order of G by n . Since G is nontrivial, $n \geq 2$. Every vertex v of G satisfies $\deg v \in [0, n-1]$ (and since G is nontrivial, note that $0 \neq n-1$). Then ...

Now fill in the rest of the proof! How many different degrees are there?

1.6: Hint: Construct a graph G in which each person is a vertex and two vertices are joined by an edge if the people they represent shake hands. The number of hands a person shakes is the degree of its vertex. Now think along the same lines as Question 1.5. What is the order of the graph? What are the possible degrees of its vertices? What can you say about where the edges are?

You're asked to find the answer to a fairly general question (i.e., where you have an arbitrary graph G and arbitrary integer r). In these situations, it's frequently helpful to start with a specific example. This won't prove the result you're after, but it might give you some insight into the problem and help you think about how you would solve the general question. For example, let M be the Martini Glass, which has $\Delta = 3$.

- 1.7:
- Can you find a 3-regular graph H_3 that contains M as an induced subgraph?
 - Can you find a 4-regular graph H_4 that contains M as an induced subgraph?
 - Can you find a 5-regular graph H_5 that contains M as an induced subgraph?
 - Look for patterns in your answers above. Can you use this information to solve the problem for the Martini Glass M for an arbitrary integer $r \geq 3$?
 - If you can do it for M , can you do it in general for an unspecified graph G ? Once again, look at what you did for the specific examples above and try to spot things that would work for an arbitrary graph G and integer $r \geq \Delta(G)$.

- 1.8 (a) Suppose that G and H are isomorphic. Then there exists an isomorphism $f : V(G) \rightarrow V(H)$. Let $v \in V(G)$. We claim that $f(N(v)) = N(f(v))$: If $u \in f(N(v))$, then there exists $u' \in N(v)$ such that $u = f(u')$. Since $vu' \in E(G)$, we have $f(v)f(u') = f(v)u \in E(H)$, i.e., $u \in N(f(v))$. This proves that $f(N(v)) \subseteq N(f(v))$. A similar proof shows that $N(f(v)) \subseteq f(N(v))$. Together, this proves that $f(N(v)) = N(f(v))$, so that $|f(N(v))| = |N(f(v))|$. But $|N(f(v))| = \deg_H f(v)$, while since f is a bijection, we have $\deg_G v = |N(v)| = |f(N(v))|$. Thus for all $v \in V(G)$ we have $\deg_G v = \deg_H f(v)$, implying that G and H have the same degree sequence.

- (b) This is false (for example, take $G = C_6$ and $H = 2C_3$).

1.9(a): Not graphical. *Why?*

We use the Erdős-Gallai Theorem. Note that

$$\sum_{i=1}^6 d_i = 4 + 4 + 3 + 2 + 1 + 0 = 14$$

which is even, so this sequence satisfies the first criterion. Now we check for each $k \in [1, 5]$ whether

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^6 \min\{k, d_i\}.$$

If the sequence passes the test for all these values of k , then it's graphical, but if there exists a k where it fails, then the sequence is not graphical.

- 1.9(b):
- For $k = 1$, we have $\sum_{i=1}^1 d_i = d_1 = 4$, while $1(1-1) + \sum_{i=2}^6 \min\{1, d_i\} = 1+1+1+1+0 = 4$, so the condition is satisfied.
 - For $k = 2$, we have $\sum_{i=1}^2 d_i = d_1 + d_2 = 4 + 4 = 8$, while $2(2-1) + \sum_{i=3}^6 \min\{2, d_i\} = 2 + 2 + 1 + 0 = 5$. The condition is not satisfied.

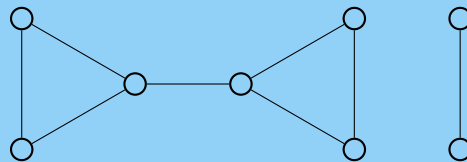
We don't need to go any further since the test fails for $k = 2$. Thus this sequence is not graphical.

1.9(c):

Repeatedly using the Havel-Hakimi Theorem, we have

$s : 3, 3, 2, 2, 2, 2, 1, 1$ is the degree sequence of a graph
if and only if $s_1 : 2, 1, 1, 2, 2, 1, 1$ is the degree sequence of a graph
if and only if $s'_1 : 2, 2, 2, 1, 1, 1, 1$ is the degree sequence of a graph
if and only if $s_2 : 1, 1, 1, 1, 1, 1$ is the degree sequence of a graph.

The sequence s_2 is the degree sequence of the graph $3K_2$. The original sequence s is therefore the degree sequence of a graph. A graph G (obtained by doing Havel-Hakimi backwards) with degree sequence s is shown below.



1.10 We discussed this question briefly in class.

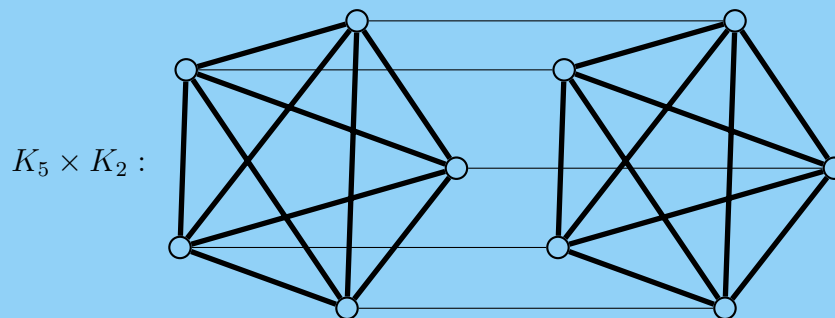
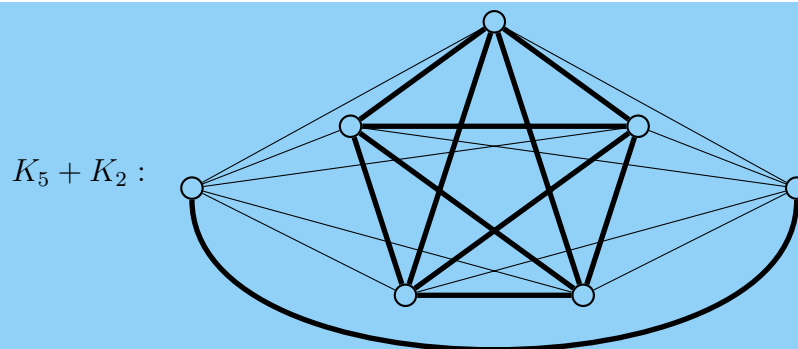
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A ‘First Theorem of Digraph Theory’ might be the following:

Theorem. If D is a digraph with m arcs, then

$$\sum_{v \in V(D)} \text{id } v = \sum_{v \in V(D)} \text{od } v = m.$$

Now prove it! Hint: Look at the proof of Theorem 1 and see how it changes when the object under consideration is a digraph rather than a graph.



Since $G \cong \overline{G}$, by a previous homework question the graphs G and $\overline{G}$ have the same degree sequence and are thus the same size (why?), which we denote m . Thus $m = n(n-1)/4$ (why?). Now consider the possible values of n so that m is an integer.

You’ll need to do some counting to answer this question. Make sure you’re familiar with the material in Appendix A of the GT notes before diving in.